SafeRoute

A shield for migrant job seekers, built on Gemini and Cloud Run

Check any job offer, chat, link or screenshot in seconds. Get a colour verdict and a spoken warning in English, Bahasa, Tagalog or Burmese.

Google Cloud AI Builder Cup 2026 | Track: Sustainability & Social Impact

Challenge theme and how we answer it

Track: Sustainability & Social Impact

Theme: **supporting communities**.

SafeRoute protects people who are recruited into forced labour and online fraud, and the families who wait for them at home. It works on the phone they already own, in the language they already speak.

Judging criterion	Where in this deck
Technical merit and Gen AI, 40%	Slides 7, 8, 10
Problem alignment and impact, 25%	Slides 3, 4, 10
Innovation and creativity, 25%	Slides 6, 8
UX and solution design, 10%	Slides 5, 6, 9

Vision: no one leaves home for a job that is a trap. Every offer gets checked, in the person's own language, before they say yes.

The problem: recruitment ads that lead to scam compounds

220,000+
people may be held in forced online scam work: at least 120,000 in Myanmar and 100,000 in Cambodia (OHCHR, 2023)

$88–114 bn
estimated scam losses across East and Southeast Asia, Australia and New Zealand in 2025 (UNODC, 2026)

$442 bn
lost to scams by consumers in 42 countries in one year; 57% of adults were scammed (GASA, 2025)

- Many victims answer a real-looking job ad: data entry, customer service, "no experience needed".
- They travel, then lose their passport and are made to run fraud on others.
- UNODC has identified people from at least 80 countries and territories in scam compounds in the region.
- 64% of scams end within a day of first contact (GASA). The check has to happen before that.

Sources: OHCHR via UN News, 29 Aug 2023, news.un.org/en/story/2023/08/1140187 | UNODC report of 21 Jul 2026 via OCCRP, occrp.org/en/news/un-report-details-southeast-asias-interconnected-criminal-economy | GASA and Feedzai, Global State of Scams 2025, feedzai.com/blog/gasa-global-state-of-scams-report

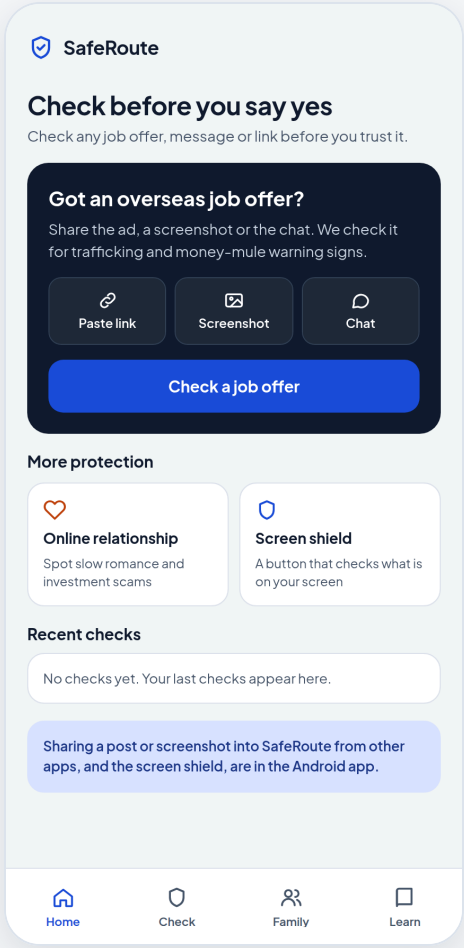
Who it is for: Rina, and the family who waits

Rina, 24, job seeker

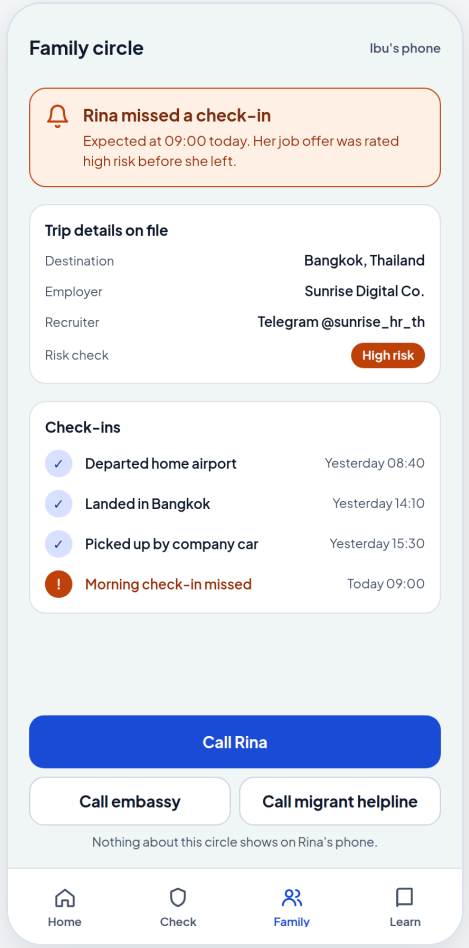
- Sees "Data entry staff, Thailand. High salary, free flight, no experience needed" on TikTok.
- Reads in Bahasa or Burmese, on a mid-range Android phone.
- Cannot judge a recruiter, a visa claim or a company registry by herself.

Ibu, her mother

- Sees Rina's check result in the Family Guardian view.
- Gets a missed check-in alert and one-tap calls to Rina, an embassy or an NGO partner.



Home



Family circle

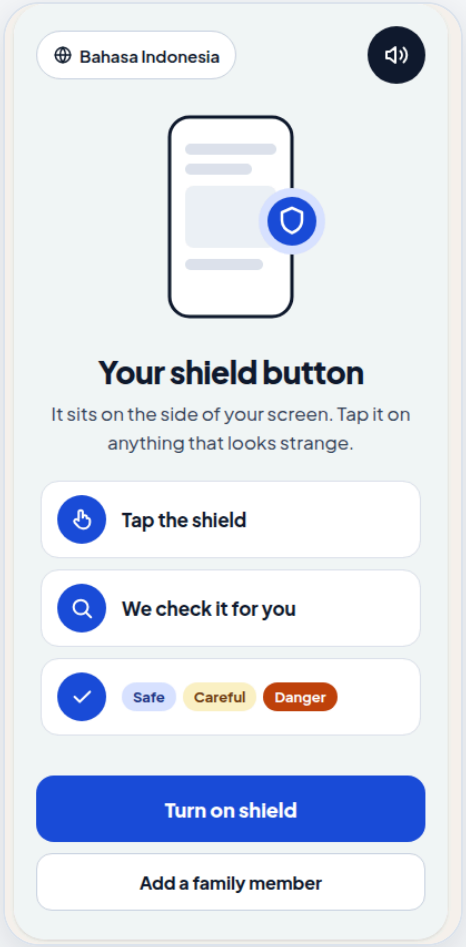
Rina and Ibu are design personas, not real users. Screens are captures of the running web app; Family data is sample data.

The solution: three shields, one button

Job Shield
Overseas offers, agency pages, recruiter claims. Checks trafficking signs, the employer and the pay.

Message Shield
Chats, SMS, links. Catches romance and pig-butchering grooming, parcel phishing and known-bad links.

Shop Shield
Marketplace listings and sellers. Flags prices that are too low and payment routes that cannot be recovered.



Floating Shield

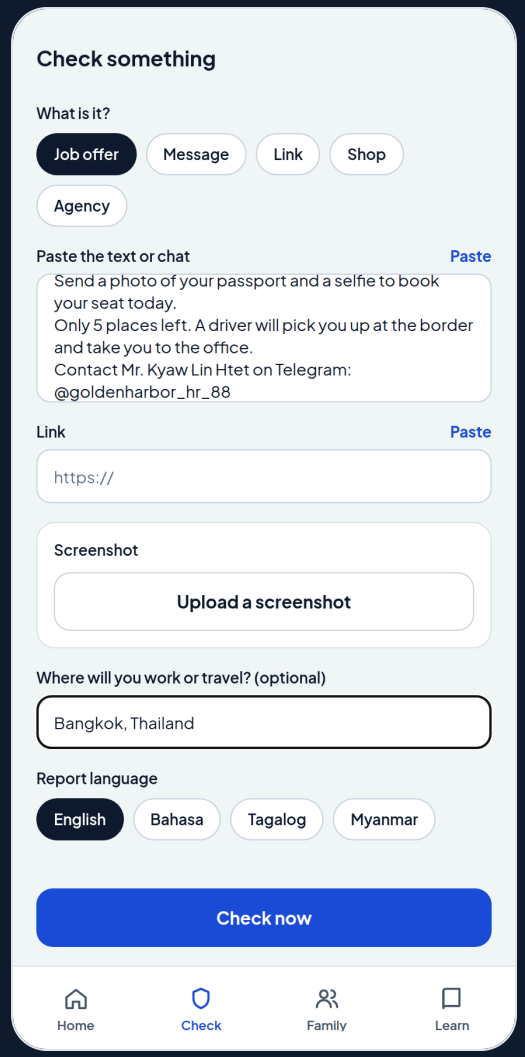
Ways in: paste text, share a link, upload a screenshot, or tap the floating Shield bubble over any app.

Ways out: a verdict in three colours (safe, careful, danger), the evidence, and a spoken warning.

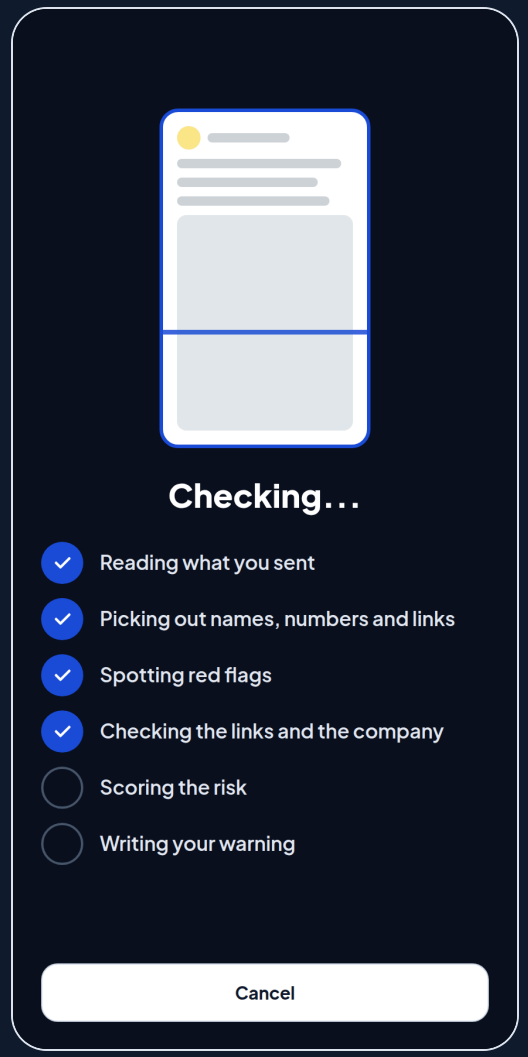
Safe Careful Danger

Built for low literacy: large buttons, one verdict word, audio for every warning, four languages.

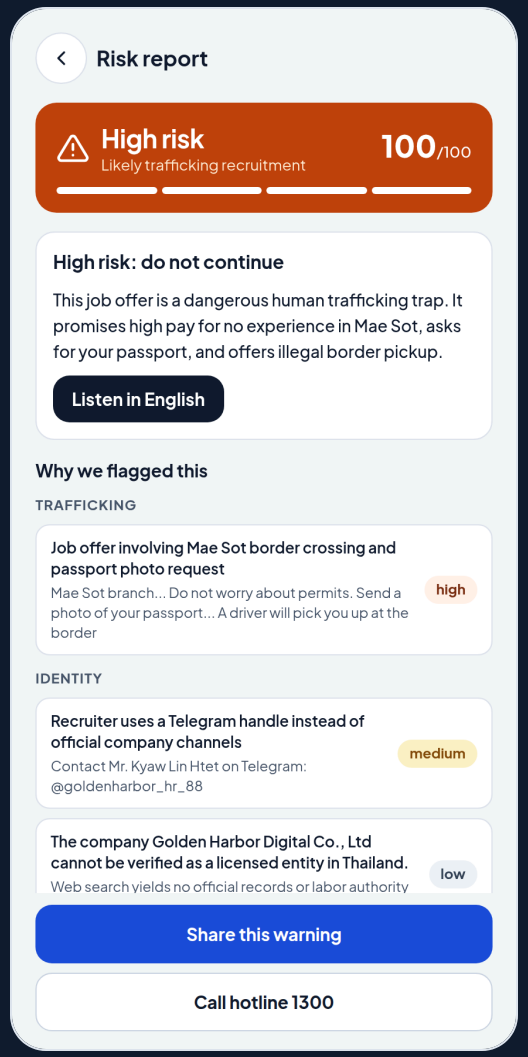
Live demo: a job ad in, a verdict out



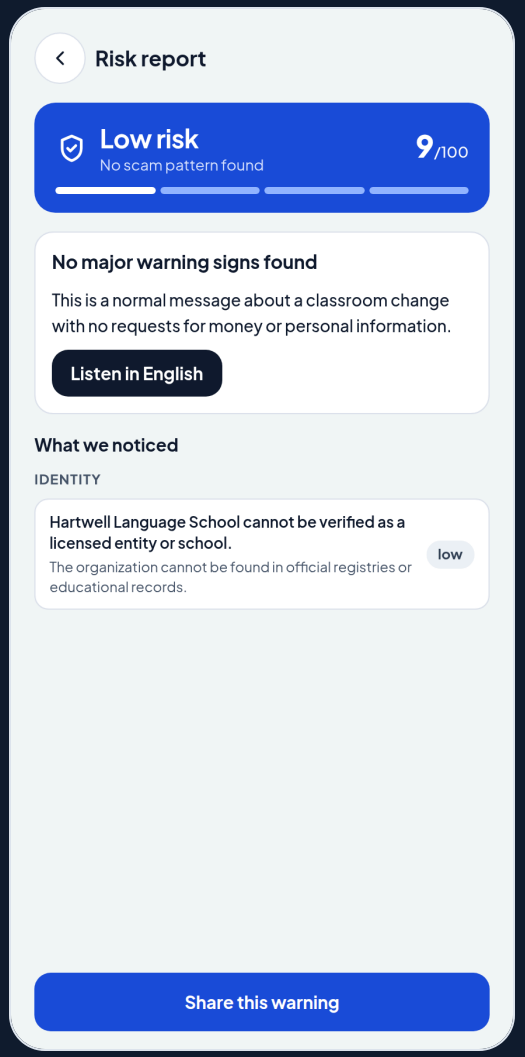
1. Paste, upload or share



2. Six stages run



3. Danger, with voice



4. Safe: no false alarm

Architecture: six stages on Google Cloud

Stage	What it does	Google service
1. Capture	App or web demo sends text, link or screenshot with language	Cloud Run, `POST /v1/check`; web on Firebase Hosting
2. Extract	One multimodal call reads text and image, returns flags in 7 families as JSON	Gemini on Vertex AI, JSON response schema
3. Detect	Verifies named recruiters, companies, phones and pay against the open web	Gemini with Grounding with Google Search
4. Correlate	Checks every URL for known threats and domain age (parallel with 3)	Web Risk API, RDAP
5. Score	Deterministic weighted rules set the score and verdict	Cloud Run, no model call
6. Explain	Plain-language verdict and spoken warning in the user's language	Gemini text-to-speech, `POST /v1/speak`

One Cloud Run service. One service account with `roles/aiplatform.user`, no stored keys. A 15 second budget per check: a late stage returns "unknown" and the verdict still ships.

Why Gen AI is essential, and what is new

Multimodal screenshots

Scam ads and chats arrive as images, with Burmese or Thai text and mixed layouts. One Gemini call reads the image and returns structured flags. Rule-based OCR cannot judge intent.

Grounding on the open web

A fake agency has no fixed blocklist entry. Gemini searches Google to check that the recruiter or company exists and is licensed, and shows the source pages it found.

Deterministic score guard

Gemini finds and explains evidence. A tested rules engine decides the verdict. The model cannot raise a result that has no evidence.

Native-language voice

The warning is spoken by Gemini text-to-speech, so Burmese works on phones with no Burmese voice installed.

Responsible AI and privacy

The model does not decide alone

- Each flag family counts once, at its highest severity.
- A Web Risk hit forces danger. A high trafficking flag forces danger.
- Grounded findings are capped at medium, so one search result never decides a verdict.
- Same input and same link results give the same score. The rules have unit tests.
- No flags means no danger, whatever the model says.

Your data stays yours

- Screenshots and text are processed in memory. The service stores no request data.
- Request text and images are never written to logs.
- No API keys on the device or in the image: the service account signs calls.
- Every verdict shows its evidence and its sources, so the user can disagree.
- Hotline numbers are configured per destination and confirmed before launch.

Feasibility, scalability and cost

- **Built and measured:** Cloud Run API, pipeline, rules engine, 49 server tests, Android app. A 19-case eval (12 scams, 7 legitimate; English, Burmese, Bahasa, Tagalog) scored 19 of 19, p50 4.4 s and p95 7.5 s per check.
- **Scales with traffic:** stateless Cloud Run autoscaling. No database to size.
- **Fails soft:** each stage has a timeout, and the verdict is built from the evidence that arrived.
- **Abuse control:** per-IP rate limits, size limits, CORS at the service edge.
- **Model choice:** gemini-3.5-flash-lite was about 2× faster than gemini-3.8-flash on the same eval.

Estimated cost per check	USD
Extract (Gemini 3.5 Flash-Lite)	0.003
Spoken warning (TTS)	0.003
Grounding, when an entity is named	0.035
Without grounding	**about 0.006**
With grounding	**about 0.04**

Estimate, not a measurement. Assumes 3,000 input and 800 output tokens, 250 audio tokens, list prices ($0.30 in, $2.50 out per 1M tokens), and grounding at $35 per 1,000 grounded prompts (assumed to apply to this model). Cloud Run compute is not included.

Pricing: ai.google.dev/gemini-api/docs/pricing. Eval: server/eval, results in server/README.md.

Roadmap

When	Milestone
Now	Android app and Cloud Run API. Web demo on Firebase Hosting for judges. 4 languages. Web Risk runs on the service account; live verification is pending.
Next 3 months	Pilot with migrant-worker NGOs and one source-country agency. Measure verdict accuracy on labelled real reports.
6 months	Real Family Guardian: check-in reminders, consented sharing, embassy and NGO case handoff.
12 months	More languages (Khmer, Lao, Thai, Vietnamese). Shared threat feed across partners. Grounded checks against national agency licence registries.

Family Guardian is design-stage today: the screens exist, the backend does not.

Team

[NAME]
[Role]

[NAME]
[Role]

[NAME]
[Role]

Web demo: [DEPLOYED URL] | Contact: [EMAIL]

Check before you say yes.